vitamins should be supplied regularly to our body. The missing numbers in B vitamins are once considered as vitamins but no longer considered as such, and the numbers that were assigned to them now form the gaps.

Table 14.2: Vitamins, their Sources, Functions and their Deficiency disease

Vitamin	Sources	Functions	Deficiency Disease
Vitamin A (Retinol)	Liver oil, Fish, Carrot, Milk, spinach and fruits such as Papaya and mango	Vision and growth	Night blindness, Xerophthalmia Keratinisation of skin
Vitamin B_1 (Thiamine)	Yeast, Milk, Cereals, Green vegetables, Liver, Pork	Co – enzyme in the form of Thiamine pyro phosphate (TPP) in glycolysis	Beri – Beri (peripheral nerve damage)
Vitamin B_2 (Riboflavin)	Soybean, Green vegetable Yeast, Egg white, Milk, Liver kidney	Co enzyme in the form of FMN (Flavin mono nucleotide)and FAD (Flavin adenine dinucleotide) in redox reactions	Cheilosis (lesions of corner of mouth, lips and tongue)
Vitamin B_3 (Niacin)	Cereals, Green leafy vegetables, Liver, Kidney	Co enzyme in the form of NAD and $NADP^+$ in redox reactions.	Pellagra (photo sensitive dermatitis)
Vitamin B_5 (Pantothenic acid)	Mushroom, Avocado, Egg yolk, Sunflower oil	Part of coenzyme A in carbohydrate protein and Fat metabolism	Inadequate growth
Vitamin B_6 (Pyridoxine)	Meat, Cereals, Milk, Whole grains, Egg.	Co enzyme in amino acid metabolism, formation of Heme in Hemoglobin	Convulsions
Vitamin B_7 (Biotin)	Liver, kidney, Milk, Egg yolk, Vegetables, Grains	Co enzyme in fatty acid Biosynthesis	Depression, Hair loss, muscle pain.
Vitamin B_9 (Folic acid)	Egg, Meat, Beet root, Leafy vegetables, Cereals, Yeast	Nucleic acid synthesis, maturation of red blood cells	Megaloblastic anaemia

XII U14-Biomolecules_New.indd 259 11-01-2021 16:38:03